# Alex Morehead

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Machine Learning • Deep Learning • Computational Biology • High-Performance Computing

## Education

**University of Missouri** Columbia, Missouri
Ph.D. Computer Science Aug 2020 - Present

- *Dissertation Advisor*: Prof. Jianlin Cheng
- *Cumulative GPA*: 4.0/4.0
- *Relevant Coursework*: Computational Intelligence, Machine Learning/Pattern Recognition, Neural Networks, Computer Vision, NLP, Unsupervised Learning, Computational Systems Biology, Design & Analysis of Algorithms

**Missouri Western State University** St. Joseph, Missouri
B.S. Computer Science Aug 2016 - May 2020

- *Minor*: Mathematics
- *Cumulative GPA*: 4.0/4.0 with General Studies Honors

## Professional Experience

**Bioinformatics & Machine Learning Lab** Columbia, Missouri
Graduate Research Assistant | Advisor: Jianlin Cheng Aug 2020 – Present

- Research and develop novel geometric deep learning algorithms applicable to computational biology
- Authored and publicly released the NSF-funded DeepInteract, a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts, along with the new Geometric Transformer
- Curated the NSF-funded DIPS-Plus, to date the largest feature-rich protein interface prediction dataset comprised of 42,000 protein complexes, and publicly released it alongside a dedicated manuscript

**EnCircle Technologies** Columbia, Missouri
Volunteer & Teaching Assistant | Manager: Jamie Bolyard Jun 2021 – Present

- Teach youth with autism spectrum disorder (ASD) technical skills for future employment and hobbies

**Altec Information Services Team** St. Joseph, Missouri (Virtual)
Software Development Intern | Managers: Dan White, Annie Martin Aug 2018 - Jan 2021

- Reduced miscommunication between service centers globally by engineering over 5 new Angular web applications
- Built more than 6 secure backend APIs with the Spring framework in an agile development setting
- Maintained the test environments of 3 in-house applications by writing LINQ queries and tuning up test databases

**NSF Research Experience for Undergraduates in Data Science of Risk and Human Activity** Indianapolis, Indiana
Undergraduate Research Assistant | Program Directors: George Mohler, Mohammad Al Hasan Jun 2019 - Aug 2019

- Authored an ensemble pipeline of 3 convolutional neural networks trained to detect gunshot sounds in the vicinity
- Deployed the pipeline to a cluster of Raspberry Pi 3 Model B+ microcomputers and evaluated its performance in real-world settings in collaboration with Indianapolis public safety officials
- Published as a manuscript and orally presented the gunshot sound detection project's results at the 2019 IEEE International Conference on Big Data

**NSF Research Experience for Undergraduates in Synthetic Biology** St. Joseph, Missouri
Undergraduate Research Assistant | Program Directors: Todd Eckdahl, Jeffrey Poet May 2018 - Aug 2018

- Investigated applications of combinatorial optimization to model lab experiments performed by synthetic biologists
- Produced Variant Sampler, a Java application for modeling the sample space of in vitro experiments
- Published results of the Variant Sampler project in the American Journal of Undergraduate Research

**Center for Academic Support** St. Joseph, Missouri
Computer Science Content Tutor | Manager: Karen Luke Feb 2017 - Aug 2018

- Tutored a total of 5 undergraduate students, 2 being from underrepresented groups
- Spent up to 1 hour with each student per week, reviewing important computer science topics such as data structures and object-oriented programming
- Designed and initiated a customized learning plan for each student's success

## Publications

### Accepted/Published

**A Region-Based Deep Learning Approach to Automated Retail Checkout** — CVPR '22
M. Shoman, A. Aboah, **A. Morehead**, Y. Duan, A. Daud, Y. Adu-Gyamfi — New Orleans, LA

**Geometric Transformers for Protein Interface Contact Prediction** — ICLR '22
**A. Morehead**, C. Chen, J. Cheng — (Virtual)

**High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function** — IEEE SuperComputing '21, St. Louis, MO
M. Gao, P. L. Andersen, **A. Morehead**, S. Mahmud, C. Chen, X. Chen, N. Giri, R. Roy, F. Quadir, T. C. Effler, R. Prout, S. Abraham, W. Elwasif, J. Skolnick, J. Cheng, A. Sedova

**Synthetic Biology Bicistronic Designs Support Gene Expression Equally Well in vitro and in vivo** — AJUR '20, Journal
O. Koucky, J. Wagner, S. Aguilera, B. Bashaw, Q. Chen, A. Eckdahl, E. Edman, P. Gomez, N. Hanlan, N. Kempf, D. Mattoon, S. McKlin, C. Mazariegos, **A. Morehead**, S. Q. Ong, A. Peterson, M. Rojas, K. Roland, K. Schildknecht, H. Seligmann, K. Slater, A. Tauchen, R. Tittor, T. Travieso, D. Urban, C. Willis, J. Zhou, N. L. Snyder, L. J. Heyer, J. L. Poet, T. T. Eckdahl, A. M. Campbell

**Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi** — IEEE BigData '19
**A. Morehead**, L. Ogden, G. Magee, R. Hosler, B. White, G. Mohler — Los Angeles, CA

### Under Review

**3D-Equivariant Graph Neural Networks for Protein Model Quality Assessment** — Bioinformatics '22, Journal
C. Chen, X. Chen, **A. Morehead**, T. Wu, J. Cheng

**Semi-Supervised Graph Learning Meets Dimensionality Reduction** — IJCNN '22
**A. Morehead**, W. Chantapakul, J. Cheng — Padua, Italy

### Preprint

**DIPS-Plus: The Enhanced Database of Interacting Protein Structures for Interface Prediction** — arXiv '21, Preprint Archive
**A. Morehead**, C. Chen, A. Sedova, J. Cheng

## Presentations

**Geometric Transformers for Protein Interface Contact Prediction** — Shandong University '22
**A. Morehead** — Qingdao, Shandong Province, China (Virtual)

**High-Performance Toolbox for Deep Learning of Protein Structure and Function** — IEEE SuperComputing '21
**A. Morehead**, A. Sedova, M. Gao — St. Louis, Missouri (Virtual)

**Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi** — IEEE BigData '19
**A. Morehead**, L. Ogden, G. Magee — Los Angeles, California

**Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi** — IUPUI Student Summer Poster Symposium
**A. Morehead**, L. Ogden, G. Magee — Indianapolis, Indiana

**Variant Sampling in vitro with a Scheduling Twist** — Alpha Chi National Convention '19
**A. Morehead** — Cleveland, Ohio

**Variant Sampling in vitro with a Scheduling Twist** — MWSU PORTAL Summer Research Showcase '18
**A. Morehead** — St. Joseph, Missouri

**Predicting Game Genres by Analyzing Code Structure** — CSCC Central Plains Conference '18
S. Frazier, **A. Morehead**, S. Prine — Maryville, Missouri

**Predicting Game Genres by Analyzing Code Structure** — MWSU Multidisciplinary Research Day '18
S. Frazier, **A. Morehead**, S. Prine — St. Joseph, Missouri

## Leadership Activities

**ICLR Volunteer** Apr 2022
**NeurIPS Reviewer** Sep 2021 – Present
**Upsilon Pi Epsilon (ΥΠΕ) | University of Missouri Chapter** Aug 2020 - Present
**Alpha Chi (AX) | Missouri Western State University Chapter** Mar 2018 – May 2020
**Kappa Mu Epsilon (KME) | Missouri Western State University Chapter** Mar 2018 – May 2020

## Awards & Grants

**Dean's Engineering Excellence Fellowship** *Aug 2020*
**James W. and Joan M. O'Neill Graduate Fellowship in Engineering** *Aug 2020*
**MWSU Outstanding Graduating Computer Science Student Award** *May 2020*
**MWSU President's Honor Roll** *May 2020*
**Floyd Tesmer/Strayer University Prize in Computer Science and Engineering** *Apr 2019*
**Alpha Chi Region IV Scholarship** *Apr 2019*
**Grand Midwest Asynchronous Programming Contest 3rd Place Prize** *Apr 2017*
**East Side Lions Club Scholarship** *May 2016*

## Projects

**BioPython (3.1k Stars) | GitHub:** https://github.com/biopython/biopython Apr 2022 – Present
- Created functionality to remove large feature generation temporary files after all processing completes
- Collaborated with senior project developers to ensure proposed solution matched existing code standards

**Graphein (300+ Stars) | GitHub:** https://github.com/a-r-j/graphein Apr 2022 – Present
- Added workflow to catch illusive OS errors when executing external programs
- Worked with senior project developer to ensure workflow matched existing code standards

**EGNN-PyTorch (150+ Stars) | GitHub:** https://github.com/lucidrains/egnn-pytorch Apr 2021 – Present
- Noticed the omission of an important geometric unit test for this neural network architecture
- Developed and validated the code to test for node-wise permutation equivariance within the network

**DeepInteract (20+ Stars) | GitHub:** https://github.com/BioinfoMachineLearning/DeepInteract Oct 2021 – Present
- Designed and developed a state-of-the-art geometric deep learning pipeline for predicting protein interface contacts
- Maintain project by addressing bug fixes, feature requests, and future planning

**DIPS-Plus (15+ Stars) | GitHub:** https://github.com/BioinfoMachineLearning/DIPS-Plus June 2021 – Present
- Curated (to date) the largest molecular dataset for machine learning of protein interface contacts
- Work to ensure ease of use and adoption by responding to feature requests and open issues

## Technical Skills

**Programming**: Python 3 • Java • Angular 2+ • C/C++ • HTML/CSS • SQL • C# • R • MATLAB

**Tools/Frameworks**: PyTorch • TensorFlow • NumPy • Pandas • Spring Boot • NodeJS • Docker • AWS • Google Cloud

**Methodologies**: RESTFUL APIs • Version Control {Git} • Agile Development

## News & Media Outreach

[1] MU Engineering. (2020, October 27). *Alex Morehead*. University of Missouri College of Engineering. https://engineering.missouri.edu/2020/10/alex-morehead/.

[2] MU Engineering. (2021, October 18). *New protein prediction tool could accelerate biological discoveries*. University of Missouri College of Engineering. https://engineering.missouri.edu/2021/10/new-protein-prediction-tool-could-accelerate-biological-discoveries/.

[3] ORNL. (2021, January 10). *Summit Powers Novel Protein Function Prediction Work*. Oak Ridge National Laboratory. https://www.ornl.gov/news/scientists-use-summit-supercomputer-deep-learning-predict-protein-functions-genome-scale/.

[4] HPCwire. (2021, January 13). *Summit Powers Novel Protein Function Prediction Work*. HPCwire. https://www.hpcwire.com/2022/01/13/summit-powers-novel-protein-function-prediction-work/.

[5] MarkTechPost. (2022, January 18). *Researchers Introduce High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function*. MarkTechPost Tech News – AI Paper Summary. https://www.marktechpost.com/2022/01/18/researchers-introduce-high-performance-deep-learning-toolbox-for-genome-scale-prediction-of-protein-structure-and-function/.